Business Finance Midterm

Written Calculations · Chapters 2–7 · 15 questions

Exhibit A · Harborview Outfitters, Inc.

Used in questions 1–3

Balance sheets, December 31 ($ millions)

Item	2024	2025
Assets		
Cash	150	200
Accounts receivable	600	700
Inventory	850	900
Total current assets	1,600	1,800
Net fixed assets	4,400	4,600
Total assets	6,000	6,400
Liabilities and owners' equity		
Accounts payable	520	560
Notes payable	280	240
Total current liabilities	800	800
Long-term debt	1,800	1,900
Common stock and paid-in surplus	1,400	1,500
Retained earnings	2,000	2,200
Total owners' equity	3,400	3,700
Total liabilities and owners' equity	6,000	6,400

2025 income statement ($ millions)

Item	2025
Net sales	4,800
Cost of goods sold	2,900
Depreciation	500
Earnings before interest and taxes	1,400
Interest paid	200
Taxable income	1,200
Taxes (21%)	252
Net income	948
Dividends	748
Addition to retained earnings	200

1. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Compute the 2025 cash flow from assets in two ways: (a) from the asset side, CFFA = OCF − NCS − ΔNWC; (b) from the financing side, CFFA = CF to creditors + CF to stockholders, where net new borrowing is the change in long-term debt and net new equity is the change in common stock and paid-in surplus. Write each formula and show that both give the same figure.

2. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Using 2025 year-end figures, compute the profit margin, total asset turnover and equity multiplier, combine them into ROE with the DuPont identity, and confirm the result with NI / TE.

3. [Exhibit A] Harborview Outfitters, Inc. ($ millions). For 2025, compute the current ratio, the quick ratio, the total debt ratio and the debt-equity ratio. Write each formula.

4. Assume the tax schedule in the table. A taxpayer reports taxable income of $200,000. Compute the total tax, the average tax rate and the marginal tax rate.

Tax schedule

Taxable income	Rate
Up to $50,000	10%
$50,001 – $150,000	20%
Above $150,000	30%

5. An investment of $12,000 today will be worth $20,000 in 7 years. (a) What annual rate does it earn? (b) At that rate, how many years would it take $12,000 to grow to $30,000? Write the equation for each part before solving.

6. You deposit $3,000 today and another $5,000 in 18 months. The account pays 6%, compounded monthly. How much is in the account at the end of 4 years? Show the value of each deposit separately.

7. You borrow $12,000 for 3 years at 7.2%, compounded monthly, repaid in equal monthly payments. Compute (a) the monthly payment, (b) the interest in the first payment and (c) the loan balance after the 12th payment.

8. A two-year office lease requires $1,200 at the beginning of every month. The relevant rate is 7.2%, compounded monthly. Value it first as an ordinary annuity and then as the annuity due it actually is.

9. A $22,000 car can be financed with 48 monthly payments of $540. Write the equation for the monthly rate, then report the monthly rate, the quoted annual rate (APR) and the effective annual rate.

10. Lender A charges 11.8%, compounded monthly. Lender B charges 12%, compounded semiannually. Convert both quotes to effective annual rates and state which loan is cheaper.

11. A $1,000 bond pays a 6% coupon semiannually and matures in 12 years. Price it when comparable bonds yield 7% and when they yield 5% (quoted annual rates), and say whether it sells at a premium or a discount in each case.

12. A $1,000 bond pays a 5% coupon semiannually, matures in 8 years and trades at $940. Write the pricing equation, then find the six-month yield, the quoted YTM, the effective annual yield and the current yield, and rank the coupon rate, current yield and YTM.

13. A company just paid a dividend of $1.80 per share. Dividends grow at 4% a year forever and investors require 12%. Compute (a) today's price, (b) the expected price three years from today and (c) the dividend yield and capital gains yield.

14. A firm just paid a $1.50 dividend. Dividends will grow at 15% for the next two years and at 4% forever after that. The required return is 12%. Write out each dividend, the terminal price and the value today.

15. A firm will pay no dividend for the next three years. It will pay $1.00 at the end of year 4, and that dividend will then grow at 5% forever. The required return is 15%. Value the stock today, showing the date at which you apply the growth model.

Business Finance Midterm — Model answers

1. Ch.2 · Cash flow from assets, two ways
OCF = EBIT + depreciation − taxes = 1,400 + 500 − 252 = 1,648
NCS = ending net FA − beginning net FA + depreciation = 4,600 − 4,400 + 500 = 700
ΔNWC = (1,800 − 800) − (1,600 − 800) = 1,000 − 800 = 200
CFFA = OCF − NCS − ΔNWC = 1,648 − 700 − 200 = 748
CF to creditors = interest paid − net new borrowing = 200 − (1,900 − 1,800) = 100
CF to stockholders = dividends paid − net new equity = 748 − (1,500 − 1,400) = 648
Check = 100 + 648 = 748, the same as the asset-side figure
Key results: 1,648 · 700 · 748 · 100 · 648

2. Ch.3 · DuPont identity
PM = NI / sales = 948 / 4,800 = 19.75%
TAT = sales / TA = 4,800 / 6,400 = 0.75
EM = TA / TE = 6,400 / 3,700 = 1.73
ROE = PM × TAT × EM = 0.1975 × 0.75 × 1.73 = 25.62%
Check = NI / TE = 948 / 3,700 = 25.62%
Key results: 19.75 · 0.75 · 1.73 · 25.62

3. Ch.3 · Liquidity and leverage
Current ratio = CA / CL = 1,800 / 800 = 2.25
Quick ratio = (CA − inventory) / CL = (1,800 − 900) / 800 = 1.13
Total debt = TA − TE = 6,400 − 3,700 = 2,700
Total debt ratio = TD / TA = 2,700 / 6,400 = 0.42
D/E = TD / TE = 2,700 / 3,700 = 0.73
Key results: 2.25 · 1.13 · 0.42 · 0.73

4. Ch.2 · Average and marginal tax rates
Tax = 0.10 × 50,000 + 0.20 × 100,000 + 0.30 × 50,000 = 5,000 + 20,000 + 15,000 = 40,000
Average rate = total tax / taxable income = 40,000 / 200,000 = 20%
Marginal rate = rate on the next dollar of income = 30%
Key results: 40,000 · 20% · 30%

5. Ch.4 · Rate and time
(a) Equation: $20{,}000 = 12{,}000 \times (1 + r)^{7}$
r = $(20{,}000 / 12{,}000)^{1/7} - 1 = 7.57\%$
BA II Plus: 7 N, −12000 PV, 0 PMT, 20000 FV, CPT I/Y = 7.57
(b) Equation: $30{,}000 = 12{,}000 \times (1.0757)^{t}$
t = ln(2.5) / ln(1.0757) = 12.56 years
BA II Plus: 7.57 I/Y, −12000 PV, 0 PMT, 30000 FV, CPT N = 12.56
Key results: 7.57 · 12.56

6. Ch.5 · Two deposits
r = 6% / 12 = 0.5% per month
FV = $3{,}000 \times (1.005)^{48} + 5{,}000 \times (1.005)^{30}$
First deposit = $3{,}000 \times (1.005)^{48} = 3{,}811.47$ (48 months)
Second deposit = $5{,}000 \times (1.005)^{30} = 5{,}807.00$ (30 months)
Total = 9,618.47
BA II Plus: 48 N, 0.5 I/Y, 0 PMT, −3000 PV, CPT FV = 3,811.47
30 N, −5000 PV, CPT FV = 5,807.00
Key results: 3,811.47 · 5,807.00 · 9,618.47

7. Ch.5 · Amortizing loan
r = 7.2% / 12 = 0.6% per month, N = 36
Payment: $12{,}000 = C \times [1 - 1/(1.006)^{36}] / 0.006$, so C = 371.62
First interest = 0.006 × 12,000 = 72.00
Balance after 12 = PV of the 24 remaining payments = $371.62 \times [1 - 1/(1.006)^{24}] / 0.006$ = 8,283.46
BA II Plus: 36 N, 0.6 I/Y, 12000 PV, 0 FV, CPT PMT = −371.62
AMORT: P1 = 1, P2 = 12, BAL = 8,283.46
Key results: 371.62 · 72.00 · 8,283.46

8. Ch.5 · Lease value (annuity due)
r = 0.6% per month, N = 24
Ordinary annuity = $1{,}200 \times [1 - 1/(1.006)^{24}] / 0.006$ = 26,747.92
Annuity due = ordinary annuity × (1 + r) = 26,747.92 × 1.006 = 26,908.41
BA II Plus: 24 N, 0.6 I/Y, −1200 PMT, 0 FV, CPT PV = 26,747.92, then × 1.006 (or BGN mode, switched off afterwards)
Key results: 26,747.92 · 26,908.41

9. Ch.5 · Rate on a car loan
Formula: $22{,}000 = 540 \times [1 - 1/(1 + r)^{48}] / r$, solved by iteration
Monthly rate = 0.6902%
APR = 0.6902% × 12 = 8.28%
EAR = $(1.006902)^{12} - 1$ = 8.60%
BA II Plus: 48 N, 22000 PV, −540 PMT, 0 FV, CPT I/Y = 0.6902
ICONV: NOM = 8.28, C/Y = 12, CPT EFF = 8.60
Key results: 0.69 · 8.28 · 8.60

10. Ch.5 · Comparing lenders
EAR(A) = $(1 + 0.118/12)^{12} - 1$ = 12.46%
EAR(B) = $(1 + 0.12/2)^{2} - 1$ = 12.36%
Decision: borrow from Lender B, which has the lower EAR
ICONV: NOM = 11.8, C/Y = 12, CPT EFF = 12.46
NOM = 12, C/Y = 2, CPT EFF = 12.36
Key results: 12.46 · 12.36 · Lender B

11. Ch.6 · Bond pricing
Inputs: N = 12 × 2 = 24, PMT = 60 / 2 = 30, FV = 1,000
At 7%: $P = 30 \times [1 - 1/(1.035)^{24}] / 0.035 + 1{,}000 / (1.035)^{24}$ = 919.71, a discount (coupon rate < yield)
At 5%: $P = 30 \times [1 - 1/(1.025)^{24}] / 0.025 + 1{,}000 / (1.025)^{24}$ = 1,089.42, a premium (coupon rate > yield)
BA II Plus: 24 N, 3.5 I/Y, 30 PMT, 1000 FV, CPT PV = −919.71
2.5 I/Y, CPT PV = −1,089.42
Key results: 919.71 · 1,089.42 · discount · premium

12. Ch.6 · Yield to maturity
Formula: $940 = 25 \times [1 - 1/(1 + r)^{16}] / r + 1{,}000 / (1 + r)^{16}$, solved by iteration
Six-month yield = 2.98%
Quoted YTM = 2.98% × 2 = 5.95%
Effective annual yield = $(1.0298)^{2} - 1$ = 6.04%
Current yield = annual coupon / price = 50 / 940 = 5.32%
Ranking: coupon rate 5.00% < current yield 5.32% < YTM 5.95% (a discount bond)
BA II Plus: 16 N, −940 PV, 25 PMT, 1000 FV, CPT I/Y = 2.98
Key results: 2.98 · 5.95 · 6.04 · 5.32

13. Ch.7 · Constant growth
$D_1 = D_0 \times (1 + g) = 1.80 \times 1.04 = 1.872$
$P_0 = D_1 / (R - g) = 1.872 / 0.08 = 23.40$
$P_3 = D_4 / (R - g) = 1.80 \times (1.04)^4 / 0.08 = 2.1057 / 0.08 = 26.32$
Dividend yield = $D_1 / P_0 = 1.872 / 23.40 = 8\%$
Capital gains yield = g = 4% (R = 8% + 4% = 12%)
Key results: 23.40 · 26.32 · 8% · 4%

14. Ch.7 · Nonconstant growth
$D_1 = 1.50 \times 1.15 = 1.725$
$D_2 = 1.725 \times 1.15 = 1.98375$
$D_3 = 1.98375 \times 1.04 = 2.0631$
$P_2 = D_3 / (R - g) = 2.0631 / (0.12 - 0.04) = 25.79$
$P_0 = 1.725 / 1.12 + (1.98375 + 25.79) / (1.12)^2 = 1.540 + 22.140 = 23.68$
BA II Plus CF: CF0 = 0, C01 = 1.725, C02 = 27.773, NPV I = 12, CPT = 23.68
Key results: 1.725 · 1.98375 · 2.0631 · 25.79 · 23.68

15. Ch.7 · A firm that pays nothing yet
Growth model at date 3, one period before the first dividend: $P_3 = D_4 / (R - g) = 1.00 / (0.15 - 0.05) = 10.00$
$P_0 = P_3 / (1.15)^3 = 10.00 / 1.5209 = 6.58$
BA II Plus: 3 N, 15 I/Y, 0 PMT, 10 FV, CPT PV = −6.58
Key results: 10.00 · 6.58